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B.Tech. Degree VI Semester Special Supplementary Examination

January 2017

ME 1601 DYNAMICS OF MACHINERY (2012 scheme)

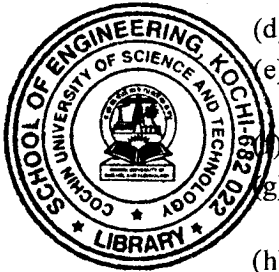
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain the superposition of solutions in the static force analysis of a four bar mechanism.
- (b) Derive an expression for the inertia force in the reciprocating parts, neglecting the weight of the connecting rod.
- (c) Explain the terms 'fluctuation of energy' and 'fluctuation of speed' as applied to flywheels.
- (d) Derive an expression for the braking torque in an internal expanding shoe brake.
- (e) Explain the effects of partial balancing of reciprocating parts in two cylinder locomotives.
- (f) Explain the principle of working of balancing machines.
- (g) Derive an expression for the ratio of belt tensions on tight and slack sides for V-belts.
- (h) Define the following terms relating to governors : (i) Hunting (ii) Sensitiveness and (iii) Isochronism.



PART B

(4 × 15 = 60)

- II. Determine the required input torque on the crank of a slider crank mechanism for the static equilibrium when the applied piston load is 1500 N. The lengths of the crank and the connecting rod are 40 mm and 100 mm respectively and the crank has turned through 45° from the inner dead centre.

OR

- III. In a four link mechanism ABCD, the link AB revolves with an angular velocity of 10 rad/s and angular acceleration of 25 rad/s² at the instant when it makes an angle of 45° with AD, the fixed link. The length of the links are : AB = CD = 800 mm, BC = 1000 mm and AD = 1500 mm. The mass of the links is 4 kg/m length. Determine the torque required to overcome the inertia forces, neglecting gravitational effects. Assume all links to be of uniform cross sections.

(P.T.O.)

- IV. The radius of gyration of a flywheel is 1 m and the fluctuation of speed is not to exceed 1% of the mean speed of the flywheel. If the mass of the flywheel is 3340 kg and the steam engine develops 150 kW at 135 rpm, then find (i) maximum fluctuation of energy and (ii) coefficient of fluctuation of energy.

OR

- V. A ship has a propeller of mass moment of inertia 2000 kg.m^2 . The propeller rotates at a speed of 360 rpm in clockwise sense looking from stern. Determine (i) gyroscopic couple and its effect when ship moves at 30 km/hr and steers to the left at a radius of 200 m (ii) maximum gyroscopic couple and its effect when ship pitches and moves up with amplitude 10° . The motion occurs with SHM and time period is 20 seconds.

- VI. Four masses are attached to a shaft at planes A, B, C and D at equal radii. The distance of the planes B, C and D from A are 40 cm, 50 cm and 120 cm respectively. The masses A, B and C are 60 kg, 45 kg and 70 kg respectively. If the system is in complete balance, determine the mass at D and the position of masses B, C and D with respect to A.

OR

- VII. A radial engine has three cylinders whose axes are spaced at angular interval of 120° . The three connecting rods are coupled directly to a single crank. The stroke is 120 mm and the length of each connecting rod is 180 mm. The mass of the reciprocating parts per cylinder is 2 kg. Determine the resultant primary and secondary forces acting on the frame of the engine when running at 2100 rpm.

- VIII. A 100 mm wide and 10 mm thick belt transmits 5 kW between two parallel shafts. The distance between the shaft centres is 1.5 m and the diameter of the smaller pulley is 440 mm. The driving and driven shafts rotate at 60 rpm and 150 rpm respectively. Determine the stress in the belt if the two pulleys are connected by (i) an open belt and (ii) a cross belt.

OR

- IX. Calculate the minimum speed of a Proell governor, which has equal arms each 200 mm and pivoted on the axis of rotation. The mass of each ball is 4 kg and the central mass on the sleeve is 20 kg. The extension arms of the lower links are each 600 mm long and parallel to the axis when the minimum radius of the ball is 100 mm.